the cost is in the fine print.

FinePrint reads Indian loan, credit card and insurance documents, finds the clauses that can cost you money, and prices each one in rupees.

CLAUSE 5 · PERSONAL LOAN

₹10,620

"...subject to a minimum of ₹5,000, will be deducted from the disbursal amount and is non-refundable."

ANJALI RAWAT · ROLL NO. 10632 PRANAV KORADIYA · ROLL NO. 10610

14 RISK TYPES **4** LANGUAGES **0.84** F1 HELD-OUT

WHY THIS EXISTS

the terms that cost money are written to be skipped.

A sanction letter, a credit card's terms or a health policy runs to pages of legal English. The lines that cost money are scattered through it, stated as a percentage of something, and seldom in the reader's first language.

- **Percentages, not rupees.** "3% plus GST" hides a five-figure fee.
- **Scattered.** Fees, penalties and rate changes sit far apart.
- **One language.** Many borrowers read Hindi or Marathi first.

ONE SAMPLE: A PERSONAL LOAN SANCTION LETTER

20

clauses in the letter

9

of them can cost the borrower money; 6 are high risk

₹10,620

the fee behind "3% of the loan amount plus GST", taken before the money arrives

THE TWO RULES FINEPRINT IS BUILT ON

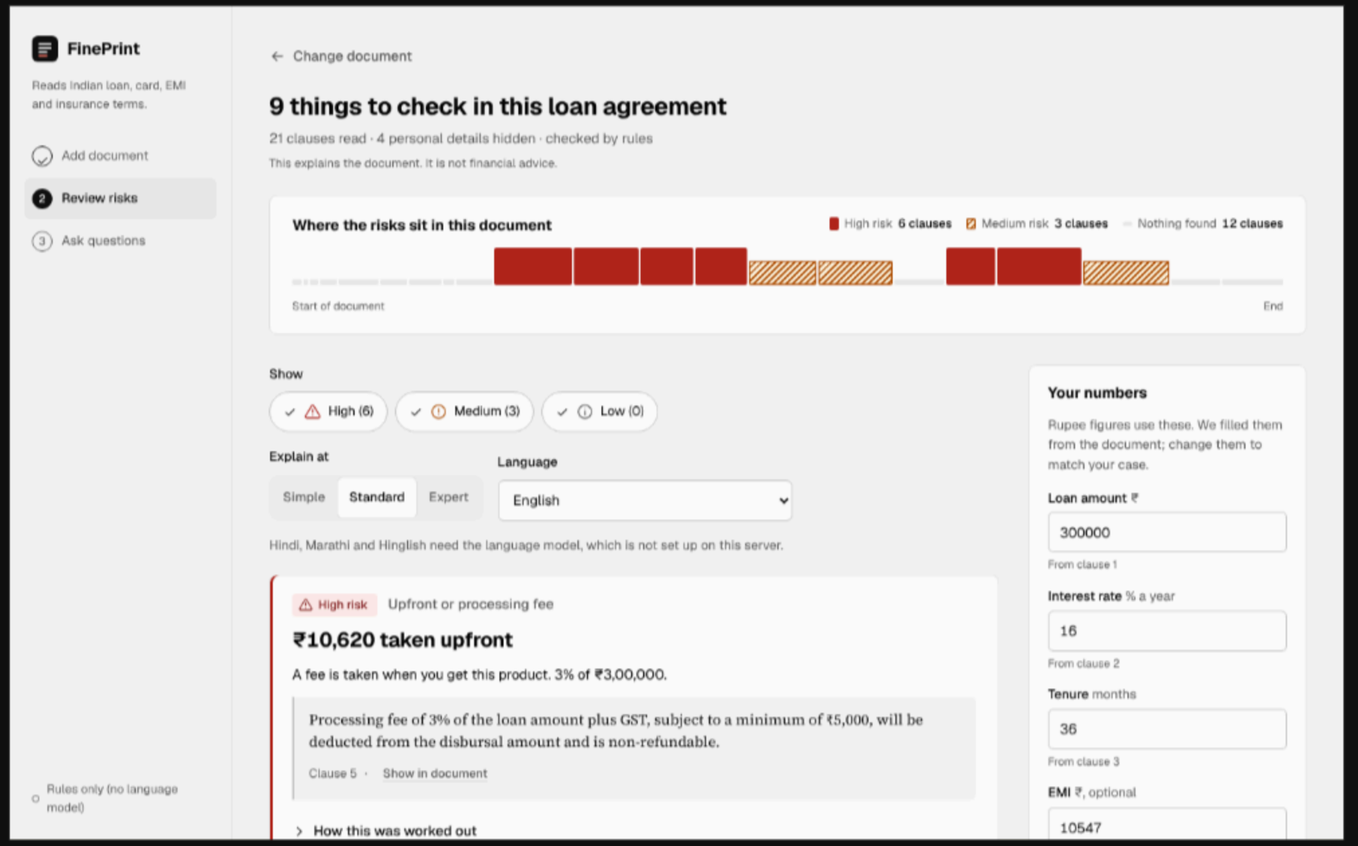
every flag quotes the document. every rupee is calculated, not generated.

A language model can help find and explain risks, but it never invents a clause and never does arithmetic. Both rules are enforced in code.

THREE STEPS

add. review. ask.

- **Add.** Paste the text, upload a PDF (bank PDFs locked against editing included) or pick one of 12 samples. Names, PAN, Aadhaar, card, account and phone numbers are hidden first.
- **Review.** Each risky clause with its severity, its cost in rupees and the exact quote. A risk map shows where in the document the problems sit.
- **Ask.** Questions in English, हिन्दी, मराठी or Hinglish, answered from the document's own clauses.



REVIEW SCREEN · SAMPLE PERSONAL LOAN · RULES ONLY, NO API KEY

THE NLP PIPELINE

six stages, one pass over the document.

01 · Ingest

read and mask

PDF or text to plain text. Personal identifiers are masked by ordered patterns, so a card number is never mistaken for Aadhaar and ₹1,50,000 is never touched.

ingest/loader.py · pii.py

02 · Segment

split into clauses

A clause starts at a numbered or bulleted line. Short title lines become headings. Every clause keeps its exact character offsets, so quotes can be checked later.

ingest/segmenter.py

03 · Extract

find the numbers

Rupee amounts with Indian commas, lakh and crore; percentages per month or year; durations. Then the loan amount, rate, tenure and EMI for the whole document.

extract/money.py · context.py

04 · Detect

flag the risks

Weighted wording cues for 14 risk categories, plus an optional language model. Every model quote must appear in the clause or the flag is dropped.

risk/rules.py · llm_classifier.py · verify.py

05 · Calculate

turn it into rupees

One deterministic function per category, fed by the clause and the document's numbers. Each result states its assumptions. The user can change the numbers.

calc/impact.py

06 · Explain & answer

say it plainly

Three reading levels, four languages. Questions are answered by BM25 retrieval over the clauses; 15 reference notes from RBI, IRDAI and general guidance say why each risk matters.

explain/ · qa/ · kb/

THE RISK TAXONOMY

14 kinds of clause that can cost you.

WHAT IT COSTS YOU · 7

- Upfront or processing fee
- Charge for paying early
- Late payment charges
- EMI bounce charge
- Annual or recurring fee
- '0% EMI' that is not free
- Paying only the minimum due

WHAT CAN CHANGE LATER · 3

- Interest rate can change
- Terms can change without your consent
- Renews and charges automatically

Each category has a default severity, wording cues with weights, and its own rupee formula. Severity can rise with the amount: a processing fee of 3% of the loan or more is high risk.

YOUR MONEY, DATA AND COVER · 4

- Money can be taken from your other accounts
- Insurance added to your loan
- Your data is shared
- Waiting period, co-pay or exclusion

RULES, MODEL, OR BOTH

rules for precision, a model for reach.

rules, always on

- **Weighted cues.** Each category has regular-expression cues with weights; a clause is flagged when the cue weight is high enough.
- **No key, no network.** Rules-only mode is the baseline every test and the evaluation run on.
- **Confidence.** A rule hit scoring 4 or more is high confidence, otherwise medium.

OPTIONAL MODEL: ANY OPENAI-COMPATIBLE API · CLAUSES SENT IN BATCHES OF 25

HYBRID: THE DEFAULT WHEN A MODEL IS SET UP

the union of both, checked.

- **Both agree:** high confidence.
- **Model only:** medium, and only if its quote is found word for word in the clause.
- **Why a union:** in a consumer-protection tool a missed trap costs more than an extra flag, and every flag quotes the document, so a reader can judge it in seconds.

FROM PERCENTAGES TO RUPEES

the language model never does arithmetic.

CLAUSE 5: "PROCESSING FEE OF 3% OF THE LOAN AMOUNT PLUS GST"

	3% × ₹3,00,000	₹9,000
+	18% GST on the fee	₹1,620
=	**taken upfront**	**₹10,620**

Plain Python, one function per category. The ₹3,00,000 came from clause 1; change it in the app and every figure updates in under a second. An explanation that used the old number is reset to the standard text.

₹8,850

if ₹1,50,000 is repaid early

CLAUSE 6 · 5% + GST

₹590

every time an EMI bounces

CLAUSE 9 · ₹500 + GST

₹211

for each month one EMI stays unpaid

CLAUSE 8 · 2% A MONTH

₹8,225

a ₹6,499 insurance premium costs once interest is paid on it

CLAUSE 10 · BUNDLED COVER

AGAINST HALLUCINATION AND OVER-REACH

four checks between the model and the reader.

Quote check

no quote, no flag

A model's flag is kept only if its quote appears in the clause, after normalising curly quotes and dashes. A paraphrase or an invented clause cannot pass.

> "N model suggestions were removed because their quote was not found in the document."

Number guard

no new numbers

An explanation may only use numbers from its clause or from the calculator. If the model changes or invents one, that explanation is replaced by the template.

> For example: the model writes "₹11,000" where the calculator said ₹10,620, so the template is shown instead.

Citation check

answers cite clauses

Answers must cite the clauses retrieved for the question. Invalid citations are removed; an answer that cites nothing is replaced by the quoted clauses.

> "...a bounce charge of ₹500 plus applicable GST..." [clause 9]

Advice guardrail

explains, never advises

Questions asking what to do, in any of the four languages, get an explanation of the terms and a plain statement that FinePrint does not give financial advice.

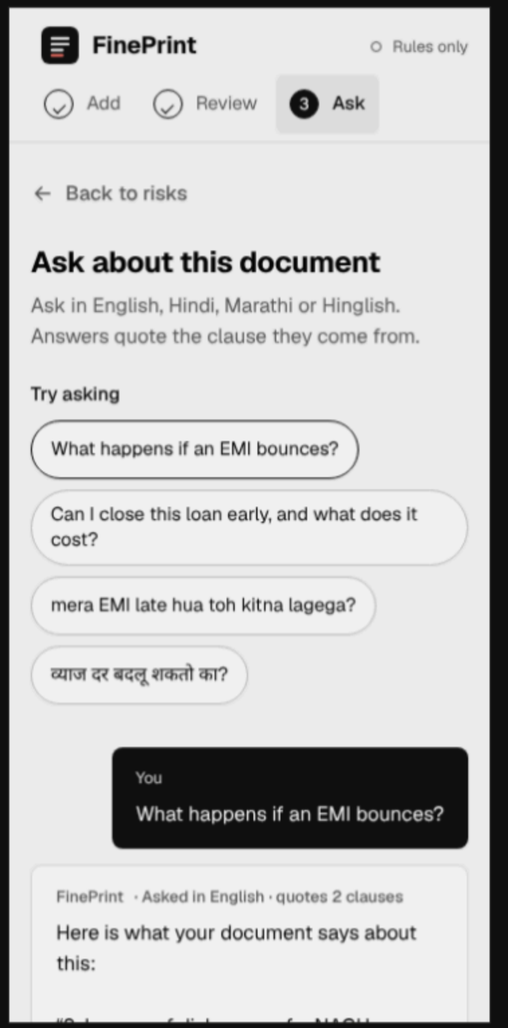
> "kya mujhe yeh loan lena chahiye?"

CODE-MIXED QUESTIONS

hinglish is not english.

Standard language detectors label Hindi typed in Roman letters as English, so retrieval fails on the Hindi words. FinePrint spots it with a function-word lexicon, then adds English synonyms to the query so BM25 can match the document.

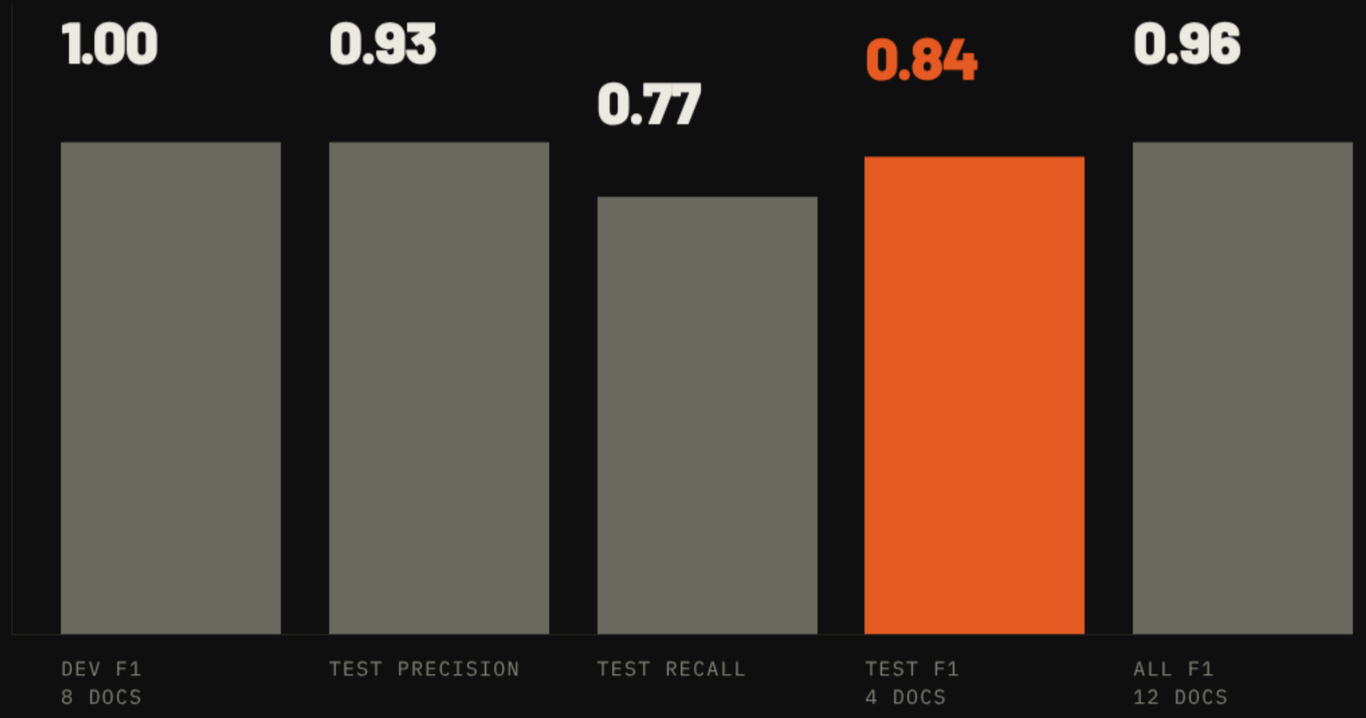
QUESTION	**mera EMI bounce hua toh kitna lagega?**
DETECTED	**Hinglish**
EXPANDED	**…** + emi instalment bounce dishonour nach ecs mandate return charges
ANSWER	**clause 9: ₹590 every time an EMI bounces · clause 8: ₹211 for each month one EMI stays unpaid**



ASK SCREEN AT 360 PX

RULES ONLY, AGAINST 64 HAND-LABELLED RISKS

0.84 F1 on documents the rules never saw.



13 / 17

test risks found, with 1 false alarm

21 / 23

pre-filled loan numbers correct (amount, rate, tenure, EMI)

88.7 → 68.2

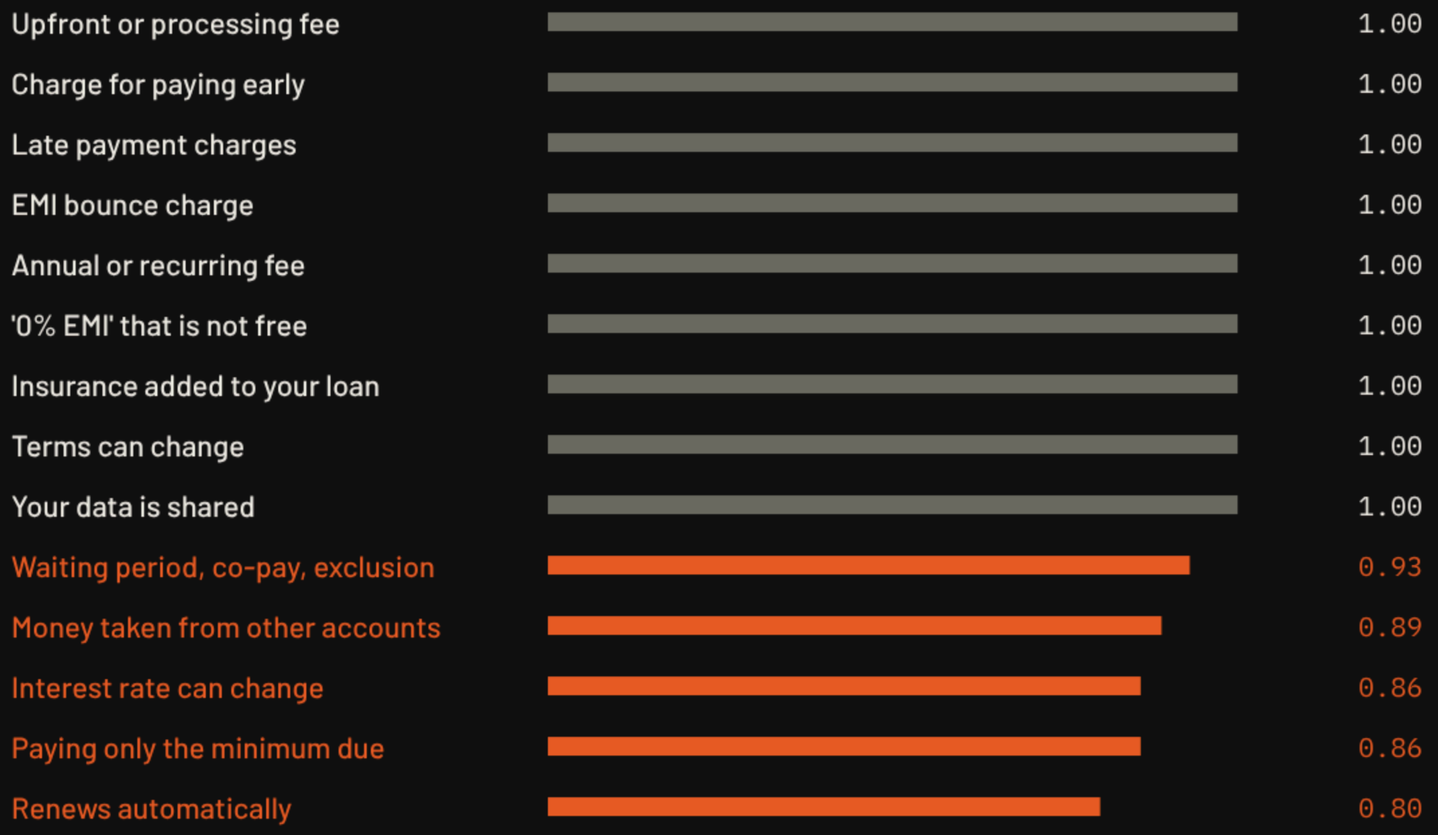
Flesch reading ease, Simple vs Expert explanations (higher is easier)

FINEPRINT

THE DEV SPLIT WAS USED TO WRITE THE RULES; THE TEST SPLIT WAS NEVER USED FOR TUNING · EVAL/RUN_EVAL.PY

F1 PER RISK CATEGORY, ALL 12 DOCUMENTS

the misses cluster in five categories.



ALL FIVE ERRORS, ALL ON TEST DOCUMENTS

MISS	"recovered from any bank account linked": set-off worded without "set-off" or "lien" (instant loan app)
MISS	"interest-free period ... is not available": the minimum-due trap stated indirectly (credit card)
MISS	"renews each year on the policy anniversary" (accident cover)
MISS	"A deductible of Rs. 5,000" (accident cover)
FALSE	"may revise premium rates at renewal" read as a floating interest rate

READING THE 0.84 CORRECTLY

what the score does not tell you.

limits today

- **Same authors.** The 12 sample documents and the rules were written by the same team, and the test split has only 4 documents. Real documents will score lower.
- **Model not yet scored.** The language-model and hybrid methods are built and tested with a stand-in model, but have not been scored with a real API key.
- **Text only.** Scanned PDFs have no text to read, and fee tables split into clauses less cleanly than sentences do.

NEXT

score it on real documents.

- **Collect.** Real sanction letters, card terms and policies in `data/real/`, kept out of git.
- **Label twice.** Two annotators label independently; agreement measured with Cohen's κ (`eval/kappa.py`).
- **Compare.** Score rules, model and hybrid on the same held-out set.

BUILT TO RUN ANYWHERE

works with no API key. stores nothing.

56
automated tests, none needing a network or a key

0
documents stored; personal data is masked before analysis

1
Docker image for Render or any host; Vercel runs it as a function

12
sample documents, each also as a PDF to test the upload

- **Stack.** Python and FastAPI on the server; plain JavaScript in the browser, no build step. Any OpenAI-compatible model (Groq, Gemini, OpenAI) through three environment variables.
- **Encrypted PDFs.** Bank PDFs locked against editing are read; a PDF that needs a password gets a clear message.
- **Every push.** CI runs the tests and the evaluation, then builds the container and smoke-tests it.
- **Usable by anyone.** 44 px touch targets, visible keyboard focus, risk never shown by colour alone, works at 360 px, back button works.

read it before you sign.

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